

Search-and-Rescue Drone System Concept

1. Core Architecture

Human defines and supervises the mission; deterministic autopilot flies the drone; machine learning identifies possible rescuees and requests inspection.

The drone uses **supervised autonomy**:

- An onboard autopilot flies and stabilizes the drone.
- A human operator defines the mission, supervises the system, and can take control at any time.
- Machine learning identifies possible rescuees but does **not** directly control the motors.
- Deterministic code applies fixed equations, thresholds, timers, safety rules, and state transitions.

2. Flight and Payload Requirements

2.1 Flight Envelope

The drone shall take off, climb, and hover with its complete rescue payload at elevations up to 2,000 m AMSL, while maintaining terrain-relative flight between 10 and 120 m AGL. It shall support a 200 m vertical climb from its launch point.

The drone should follow terrain rather than maintain one fixed altitude above sea level.

Mountain rises by 100 m

↓

Drone climbs enough to maintain 60–120 m AGL

2.2 Rescue Payload

- RGB/day camera
- IR/night camera
- Thermal camera

- 24 GHz FMCW radar
- Passive phone-RF receiver
- Microphone array
- Onboard computer for CNN, Mamba, and software processing
- Speaker
- Beacon
- Smoke-marker mechanism

2.3 Endurance Constraint

For **3–4 hours of continuous flight**, the required battery would be extremely large. A single-battery, sensor-heavy multirotor is not realistically capable of a 3–4-hour mission using conventional lithium batteries.

Average drone power	3-hour battery	4-hour battery	Approximate battery mass
1.1 kW	4.1 kWh	5.5 kWh	20–27 kg
1.5 kW	5.6 kWh	7.5 kWh	27–36 kg
2.0 kW	7.5 kWh	10 kWh	36–48 kg

3. Responsibility Partition

3.1 Human Operator

The operator:

- Defines the search area and mission objective.
- Draws the search area on a map.
- Selects the launch point, maximum altitude, route, geofence, and safety boundaries.
- Approves takeoff.
- Starts, pauses, or aborts the mission.
- Monitors the live map and sensor feeds.
- Reviews candidate-rescuee alerts.
- Makes the final confirmation that the target is likely a person.
- Approves close approach.
- Approves deployment of smoke, beacon, or other guidance equipment.
- Contacts rescue teams and shares confirmed coordinates.

- Takes manual control when required.
- Handles emergency landing, abort, and return-to-home decisions.

3.2 Deterministic Code

These functions follow fixed equations, thresholds, timers, and rules.

Flight, Navigation, and Safety

- Autonomous search-path generation
- GPS waypoint navigation
- Geofencing
- Obstacle avoidance
- Altitude control
- Battery monitoring
- Return-to-home logic
- Sensor health checks
- Fault detection
- Safety restrictions preventing lights or mirrors from pointing toward aircraft or roads

Sensor and Mission Coordination

- Day/night switching using a lux sensor
- Camera and sensor synchronization
- Speaker, strobe, mirror, and smoke-marker control
- GPS tagging of every detection
- Candidate → inspect → confirm state machine
- Detection persistence checks across several frames
- Two-sensor voting or consensus rules
- Bounding-box position matching between cameras

Audio Processing

- SOS buzzer timing
- English/Arabic announcement timing
- Silent listening-window timing
- Microphone-array beamforming
- Acoustic direction estimation
- Rotor-noise filtering using motor RPM

Passive RF Processing

- Frequency scanning
- Signal-strength measurement
- RF direction estimation from measurements taken at several drone positions

FMCW Radar Processing

- Range FFT
- Doppler FFT
- Range–Doppler map
- CFAR detection
- Range measurement
- Velocity measurement
- Phase-based micro-motion extraction

3.3 Machine Learning

These functions require trained models and annotated datasets.

Computer Vision

- Human detection in RGB images
- Human detection in IR/night-vision images
- Human detection in thermal images
- Detection of:

- Standing people

- Sitting people

- Lying people

- Partially buried people

- Partially visible body parts

- Motionless people

- Rejection of:

- Rocks

- Bushes

- Animals

- Shadows
- Hot sand
- Multi-frame visual tracking
- Person segmentation, if added later

Audio Classification

- Shouting detection
- Speech or human-voice detection
- Whistle detection
- Clapping detection
- Tapping or impact-pattern detection
- Crying detection
- Separation of human sounds from wind, animals, and rotor noise

FMCW Classification

Deterministic DSP extracts radar features. Machine learning may then classify them as:

- Human movement
- Breathing-like micro-motion
- Animal movement
- Vegetation or moving debris
- Drone-vibration interference
- Unknown target

Passive RF Classification

Machine learning may classify received bursts as:

- Cellular
- Wi-Fi
- Bluetooth
- Drone telemetry
- Background infrastructure
- Unknown RF transmission

Sensor Fusion

A model such as Mamba can combine several seconds of observations and classify the result as:

No target

Possible rescuee

Probable rescuee

High-priority rescuee

4. Guidance and Control

4.1 Deterministic Guidance Architecture

A Pixhawk-class autopilot runs the safety-critical flight controls.

Operator search area

↓

Mission planner creates search route

↓

Autopilot follows GPS waypoints

↓

IMU + GNSS + compass + barometer stabilize the drone

↓

Radar altimeter maintains height above terrain

4.2 Navigation Sensors

- **GNSS/GPS:** global position

- **IMU:** roll, pitch, yaw, and acceleration
- **Compass:** heading
- **Barometer:** approximate altitude
- **Downward radar altimeter:** actual height above snow, sand, or terrain
- **Forward/side radar:** obstacle detection in fog, darkness, and dust
- **Terrain map:** anticipation of slopes and mountains

4.3 Autonomous Mission States

1. TAKEOFF
2. CLIMB TO SEARCH HEIGHT
3. FOLLOW SEARCH PATTERN
4. DETECT POSSIBLE RESCUEE
5. SLOW DOWN AND APPROACH
6. HOVER AND VERIFY
7. ALERT OPERATOR
8. MARK LOCATION / GUIDE RESCUEE
9. CONTINUE SEARCH OR RETURN HOME

4.4 Search Patterns

- Use a lawnmower or parallel-track pattern in open desert, snowfields, and relatively flat terrain.
- Use contour-following routes in mountain areas, remaining approximately parallel to the slope.

4.5 Machine-Learning Output to the Controller

Machine learning provides candidate information rather than motor commands.

Possible person detected

Confidence: 82%

Direction: 35° right

Estimated ground position: GPS coordinate

The deterministic controller then applies the mission rules.

If detection persists:

slow down

approach candidate

maintain safe altitude

point sensors toward target

begin verification

This keeps flight control predictable when the model produces a false detection.

4.6 Candidate-Inspection Behavior

When a possible rescuee is found:

1. Record the target's estimated ground coordinates.
2. Reduce speed.
3. Descend to an authorized inspection altitude.
4. Hover or circle the location.
5. Point RGB/IR, thermal, and radar sensors toward the target.
6. Run the bilingual call-and-listen cycle.
7. Wait for sensor agreement or operator confirmation.

4.7 Communication and Manual Control

Use a dedicated encrypted command-and-control radio rather than relying solely on public cellular service.

The communication configuration may include:

- Primary long-range control radio
- Separate video/data link
- Optional LTE/5G backup where coverage exists
- Optional satellite link for remote regions
- Onboard mission storage so the drone can continue safely during brief link loss

4.8 Fail-Safe Actions

Failure	Drone response
Control link briefly lost	Hover or continue the safe route
Link not restored	Return to home
GNSS degraded	Slow down and use IMU/radar positioning
Battery reaches reserve	Stop searching and return
Dangerous wind	Abort or land at a safe point
Sensor failure	Disable the affected search mode and notify the operator
Obstacle detected	Stop, climb, or reroute
Terrain clearance uncertain	Hover and wait for the operator

5. Detection and Verification Stack

A strong rescue stack consists of:

Component	Role
Thermal camera	Primary person detection
RGB/day camera	Daytime verification
Low-light or IR camera	Night visual confirmation
24 GHz FMCW radar	Presence, range, movement, and micro-motion confirmation
Speaker and microphone array	Response prompting and acoustic confirmation
Passive RF receiver	Cellular, Wi-Fi, and Bluetooth detection
Beacon, strobe, mirror, and smoke	Attracting attention and guiding the rescuee

Operating Modes

Day:

RGB camera + thermal camera → ML person detection

Night:

Low-light/IR camera + thermal camera → ML person detection

Poor visibility:

Thermal camera + 24 GHz radar → presence confirmation

Possible rescuee found:

Hover → radar micro-motion check → camera verification → alert rescue team

6. Day, Night, and Ambient-Light Switching

Day Mode

- Thermal camera
- Daylight/RGB camera

Night Mode

- Thermal camera
- Low-light or infrared night-vision camera

Use a **light sensor as the primary trigger** and local solar time only as a backup. A time-only system could incorrectly select day mode during a sandstorm, heavy fog, dark clouds, or deep shadow.

A raw LDR can be used, but a digital lux sensor such as a **BH1750** or **VEML7700** provides more consistent measurements.

Ambient light	Mode
Above 150 lux	Daylight/RGB + thermal
50–150 lux	Transition mode: RGB + low-light/night camera + thermal
Below 50 lux	Night/low-light imaging + thermal

Hysteresis

- Switch to night mode only when ambient light remains below **50 lux for 30 seconds**.
 - Switch back to day mode only when ambient light remains above **150 lux for 30 seconds**.
-

7. Computer Vision and Dataset

Computer vision with machine learning is the **main wide-area detection system**. The 24 GHz FMCW radar acts as a confirmation sensor.

7.1 Dataset Requirements

The image dataset should represent the drone’s actual operating conditions:

- Different flight altitudes
 - Different camera angles
 - Morning, midday, sunset, and night
 - Hot and cold desert ground
 - White, black, and beige clothing
 - Partial burial in sand
 - Fog, dust, rain, and simulated reduced visibility
 - People lying completely still
 - Different body sizes and poses
-

8. FMCW Radar

FMCW radar is used for:

- Confirmation

- Close-to-medium-range human detection
- Poor-visibility support
- Micro-motion detection while the drone is stable

FMCW radar does not replace:

- Thermal imaging
- Daylight/night cameras
- Audio hailing and listening
- Passive RF or phone-based detection

Keep FMCW active during day and night. Use low-power scanning during flight and full-resolution verification during hover.

8.1 Operation by Mission Phase

Search phase	FMCW operation
Normal wide-area flight	Run continuously at a lower update rate or pulse periodically to save power
Camera or thermal candidate detected	Increase the radar update rate and focus on the candidate area
Close-range hover	Use full FMCW processing for movement and breathing confirmation
Battery-critical return flight	Disable FMCW unless it is needed for obstacle avoidance

8.2 Environmental Performance

FMCW is usually useful in:

- Night
- Haze
- Fog
- Dust
- Moderate sand
- Smoke

Detection becomes more difficult in:

- Heavy rain
- Dense wet snow

- Hail
- Very cluttered environments
- Strong drone vibration or motion
- Environments with moving vegetation or debris

Frequency-band trade-off:

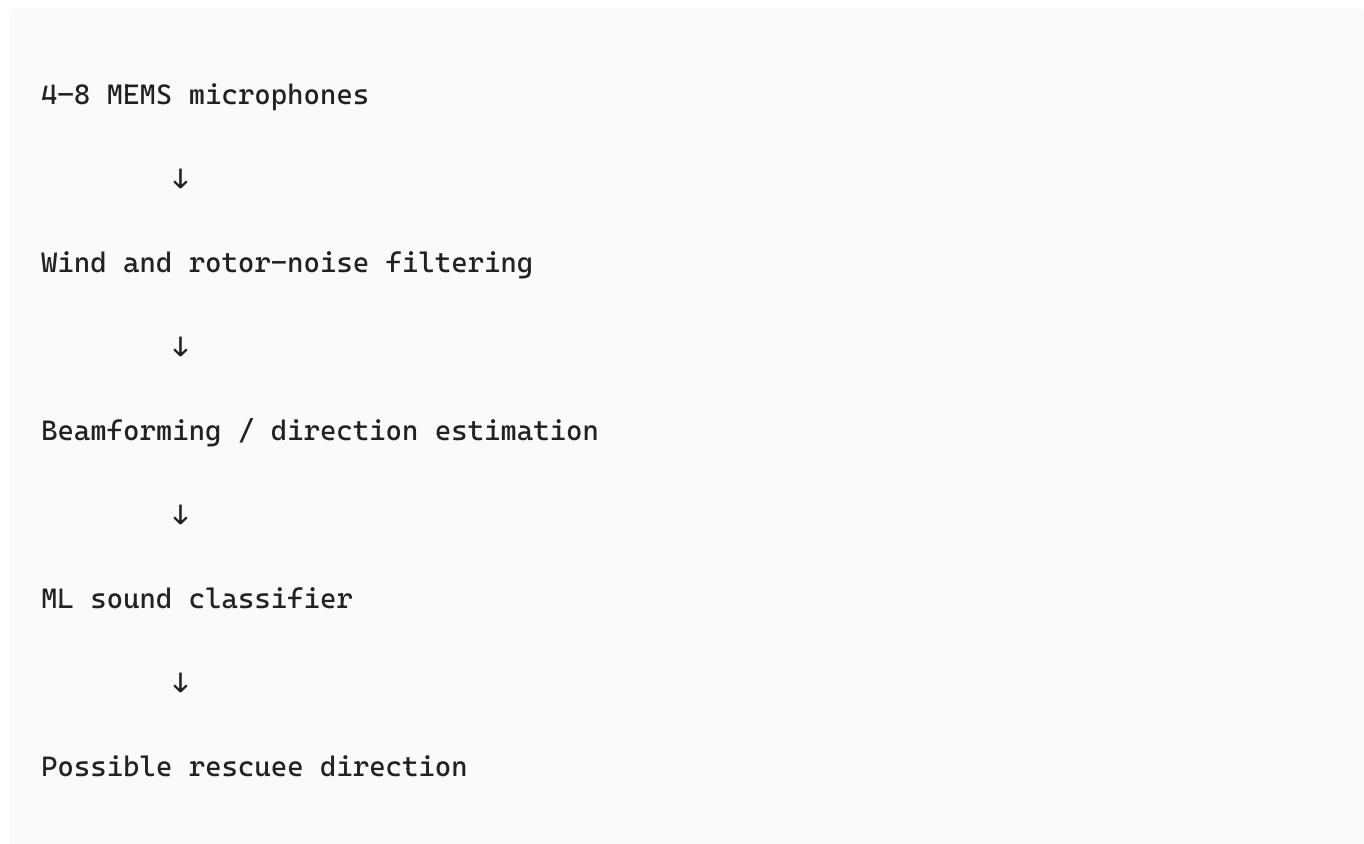
- Lower-frequency radar generally penetrates better.
- Higher-frequency mmWave provides better resolution but is more affected by atmospheric losses and clutter.

A 24 GHz FMCW radar commonly provides about **250 MHz of sweep bandwidth**. Two targets separated by less than roughly **60 cm in range** may therefore be difficult to distinguish.

9. Acoustic Detection and Call-and-Listen System

9.1 Recommended Acoustic Sensor

Use a **4–8 microphone digital MEMS array** mounted underneath the drone.



The useful speech-intelligibility range is approximately **300 Hz–5 kHz**. A practical processing band of **200 Hz–8 kHz** also captures whistles, claps, and many impact sounds.

9.2 Sound Classes

```
shouting_for_help  
  
normal_speech  
  
whistle  
  
clapping  
  
metal_or_rock_tapping  
  
crying  
  
animal  
  
wind  
  
drone_rotor_noise  
  
unknown
```

9.3 Microphone Placement

Mount the microphone array:

- Underneath the drone
- As far as practical from the motors
- Behind acoustic foam or wind protection
- On vibration-isolating rubber mounts
- With the microphones spaced around a circular plate

9.4 Listening Behavior

During the 15-second listening period:

- Turn the speaker and buzzer completely off.
- Hold the drone in position.
- Record continuously.
- Use known motor RPM for noise cancellation.
- Check for voice, whistles, claps, and tapping.

- Estimate the sound-arrival direction using the microphone array.

Create an acoustic candidate when:

- A human voice, whistle, clap, or tapping event is detected.
- Confidence is at least **70%**.
- The event appears twice during the listening period.

Then:

- Save the estimated direction.
- Turn or move toward that direction.
- Repeat the announcement and listening cycle.
- Inspect the location using thermal imaging, cameras, and FMCW radar.

9.5 SOS Sequence

Buzzer Pattern

- 3 short buzzes: **0.4 s each**, with **0.3 s gaps**
- 3 long buzzes: **1.2 s each**, with **0.3 s gaps**
- 3 short buzzes: **0.4 s each**, with **0.3 s gaps**

The complete SOS buzzer pattern lasts **8.4 seconds**.

Spoken Message

English:

“Rescue drone. Shout, whistle, clap, or hit an object now.”

Arabic:

هذه طائرة إنقاذ. اصرخ، صفّر، صفّق، أو اضرب أي جسم الآن

Cycle Timing

- SOS buzzer: **8.4 seconds**
- English and Arabic messages: **12 seconds total**
 - 6 seconds in English
 - 6 seconds in Arabic
- Silent listening period: **15 seconds**

- Complete cycle: **35.4 seconds**
- Repeat the complete cycle **3 times**

SOS buzzer



English and Arabic spoken instruction



15 seconds of silence and listening



Repeat

The silent listening window is essential because continuous buzzing or speech would mask a survivor's voice, whistle, clapping, or tapping.

During the listening period:

No buzzing, no speaker playback, and the drone flies or hovers as steadily as possible.

Reference retained from the source document:

https://www.fema.gov/pdf/emergency/usr/appen_a.pdf

10. Passive Phone-RF Detection

10.1 Phone-Associated RF Sources Considered

Source	Approximate frequencies	Typical detectable range
2G/3G/4G cellular	About 600 MHz–2.7 GHz	A few metres with a basic RF detector; hundreds of metres to several kilometres with a sensitive receiver and suitable antenna

Source	Approximate frequencies	Typical detectable range
5G sub-6 GHz	About 600 MHz–7.1 GHz	Hundreds of metres to kilometres, depending strongly on the frequency band
5G millimetre-wave on supported phones/networks	Roughly 24–40 GHz, sometimes higher depending on region	Usually tens to a few hundred metres with clear line of sight; walls, hands, and objects block it heavily
Wi-Fi	2.4 GHz, 5 GHz, and 6 GHz	Normally 20–100 m indoors or outdoors; several hundred metres with line of sight and directional equipment
Bluetooth	About 2.4 GHz	Commonly 5–30 m; sometimes around 100 m or more with Bluetooth Low Energy and good conditions
NFC	13.56 MHz	Normally 1–4 cm; specialized antennas may detect activity from tens of centimetres, but not usually metres
GPS reception	Approximately 1.2–1.6 GHz	The phone normally receives GPS and does not transmit an ordinary GPS signal to detect
Internal electronics	Weak unintended emissions from kilohertz into gigahertz	Usually centimetres to a few metres with ordinary equipment; greater distance requires specialized laboratory receivers and shielding-free conditions

10.2 Practical Detection Priorities

After filtering out weaker or impractical signals:

Source	Approximate frequencies	Practical detectable range
2G/3G/4G cellular	600 MHz–2.7 GHz	Hundreds of metres to several kilometres with suitable equipment
5G sub-6 GHz	600 MHz–7.1 GHz	Hundreds of metres to kilometres
Wi-Fi	2.4, 5, and 6 GHz	Typically 20–100 m; farther with directional antennas
Bluetooth	About 2.4 GHz	Typically 5–30 m; occasionally 100 m or more

Broad approximation order:

1. Start with 4G and earlier cellular signals.
2. Follow with sub-6 GHz 5G.
3. Use Wi-Fi and Bluetooth for closer-range approximation.

10.3 Receiver Options

Option	Coverage	Suitability
HackRF One	1 MHz–6 GHz	Lower-cost proof of concept
USRP B205mini-i	70 MHz–6 GHz	Better professional prototype; up to 56 MHz instantaneous bandwidth
AD8318 RF detector	1 MHz–8 GHz	Detects RF power only and cannot reliably identify a signal as coming from a phone

10.4 Antennas

Do not use one small omnidirectional antenna for the full frequency range. Use at least:

- A **600 MHz–2.7 GHz directional antenna** for cellular signals
- A **2.4–6 GHz directional antenna** for 5G, Wi-Fi, and Bluetooth

10.5 Detection Logic

During flight, the receiver should:

1. Scan only configured phone uplink bands.
2. Ignore stronger cellular-tower downlink bands.
3. Detect short RF bursts above the local noise level.
4. Classify the signal as cellular, Wi-Fi, Bluetooth, or unknown.
5. Record drone position, heading, and received power.
6. Compare measurements from several drone positions.
7. Estimate the likely source direction and approach cautiously.

11. Multisensor Fusion and 2D-to-1D Mapping

11.1 Recommended Approach

- Use **YOLO/CNN models** separately on RGB, IR, and thermal images.
- Use detected bounding boxes to match the same target between cameras.
- Use **Global Average Pooling embeddings** only when building a learned sensor-fusion classifier.
- Keep FMCW range, RF strength, and acoustic direction as additional one-dimensional numerical features.

Role split:

- **CNN:** detects people separately in RGB, IR, and thermal images.
- **Mamba:** combines camera, radar, audio, RF, and drone-motion information over several consecutive seconds.
- **Deterministic controller:** decides whether the drone continues searching, approaches, hovers, or reports a rescuee.

11.2 Stage 1: Per-Sensor Processing

```

RGB image      → CNN detector → boxes + confidence + feature vector

IR image       → CNN detector → boxes + confidence + feature vector

Thermal image  → CNN detector → boxes + confidence + feature vector

Radar          → range, motion, and micro-motion features

Audio          → voice, whistle, and tapping features

RF receiver    → cellular, Wi-Fi, and Bluetooth features

```

11.3 Stage 2: One Feature Vector per Time Step

```

[
    RGB_person_confidence,
    IR_person_confidence,
    thermal_person_confidence,
    target_x,

```

```
target_y,  
  
radar_range,  
  
radar_motion_score,  
  
radar_breathing_score,  
  
human_voice_score,  
  
whistle_score,  
  
phone_RF_score,  
  
drone_altitude,  
  
drone_speed  
]
```

Additional sensor-fusion inputs listed in the source include:

```
Target position  
  
Drone motion  
  
Weather and visibility
```

11.4 Stage 3: Temporal Fusion with Mamba

20–50 consecutive feature vectors

↓

Mamba

↓

No target / possible rescuee / probable rescuee / high-priority rescuee

At **10 measurements per second**, 30 feature vectors represent a **3-second observation window**.

12. Candidate and Confirmation Logic

Two-camera agreement automatically triggers inspection. One strong, persistent detection must also trigger inspection rather than being discarded.

12.1 Candidate Trigger

Create a candidate when either condition is met:

Condition A: Two-Camera Agreement

- Two active cameras detect a person at the same location.
- Confidence is at least **0.55**.
- The detection persists for at least **3 consecutive frames**.

Condition B: Strong Single-Camera Detection

- One camera detects a person.
- Confidence is at least **0.85**.
- The detection persists for at least **5 consecutive frames**.

Then:

- Slow down.
- Hover.
- Point all sensors at the candidate.
- Begin close-range verification.

12.2 Rescuee Confirmation

Confirm a probable rescuee when any of the following occurs:

- Two imaging sensors continue detecting the same target.

OR

- One imaging sensor detects the target and FMCW detects human movement or breathing-like micro-motion.

OR

- The microphone detects shouting, whistling, clapping, or tapping after the announcement.

Deterministic Example

```
If two sensors detect the same location
```

```
for at least 3 of the last 5 frames:
```

```
    create high-priority candidate
```

```
If one sensor has confidence above 0.85
```

```
for at least 5 frames:
```

```
    create inspection candidate
```

```
If imaging detection is confirmed by radar
```

```
or an audible human response:
```

```
    mark probable rescuee
```

13. LiDAR

LiDAR does not help in dense fog and should remain optional.

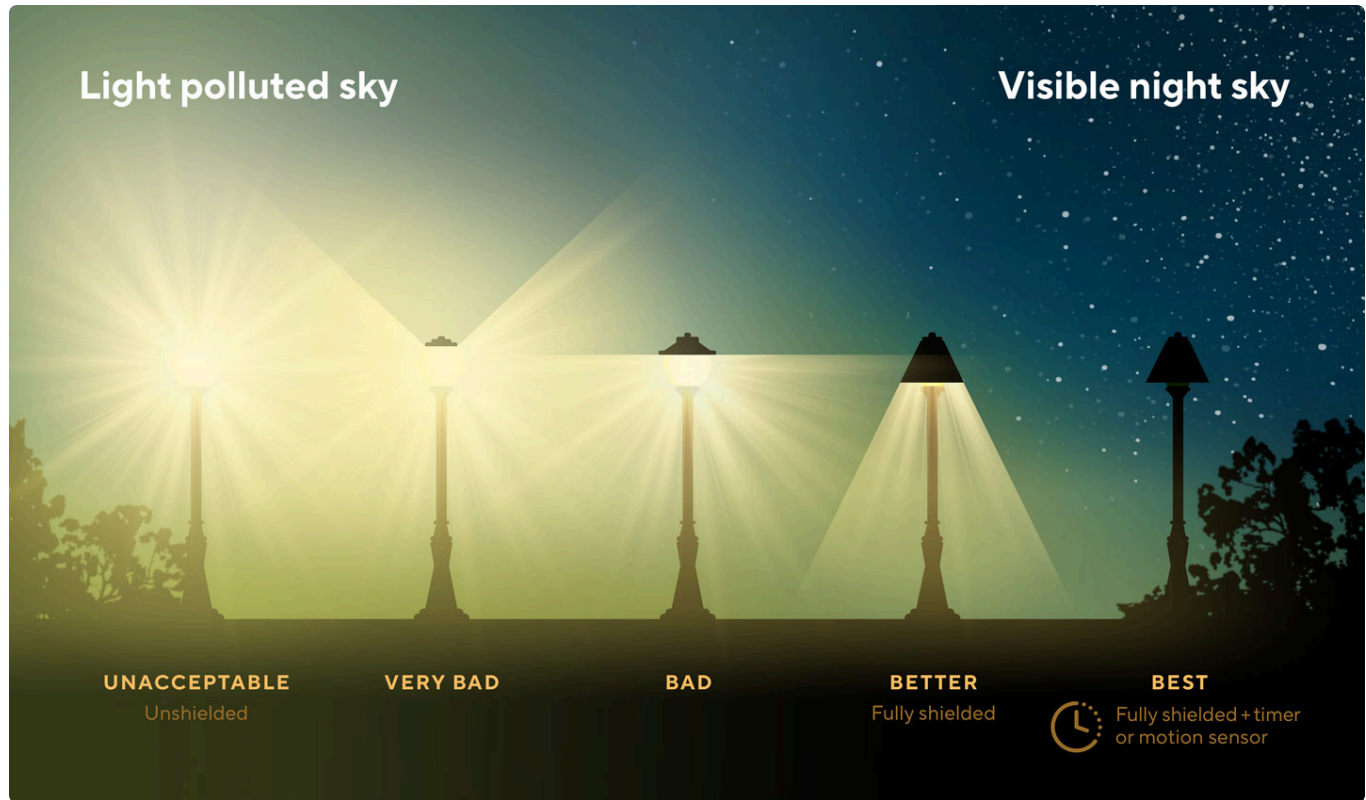
14. Visual Attention and Guidance Systems

14.1 Light Beacon Setup

- A rotating or fixed **360° amber flashing beacon** for attracting attention; the final choice between rotating and fixed remains undecided.
- A directional, long-distance, high-beam white flash operating at **2 Hz**.
- A downward white spotlight for close-range inspection.

The setup is intended to create as much light “pollution” as possible.

Original image reference:



14.2 Rotating Heliograph-Style Mirror

Mirror tilt from horizontal:

$$\text{Mirror surface tilt from horizontal} = 45 + \frac{\text{solar elevation}}{2}$$

For a representative Dubai location around **25.2° N, 55.3° E**:

Period	Noon Sun elevation	Required mirror tilt
Winter solstice	About 41.4°	About 65.7°
Equinoxes	About 64.8°	About 77.4°
Summer solstice	About 88.2°	About 89.1°

Dubai time	Sun azimuth	Sun elevation	Mirror tilt for straight down
08:00	83° , east	28°	59°
10:00	96° , east-southeast	55°	73°
12:24 solar noon	178° , south	82°	86°
14:00	255° , west-southwest	67°	78°
16:00	272° , west	40°	65°
18:00	283° , west-northwest	13°	51°

The source leaves open whether to use the calculated tilt or a rotating **45–90° range**.

Reference retained from the source document:

<https://gml.noaa.gov/grad/solcalc/azel.html>

14.3 Ground Distance Covered

For drone altitude (H), the horizontal ground distance reached by the reflected flash is:

$$R = H \tan \theta$$

For wider coverage without electronic beam steering, use three small mirrors at different tilts:

Mirror	Angle from vertical	Purpose
Mirror 1	40°	Near-range flash
Mirror 2	60°	Medium-range flash
Mirror 3	75°	Far-range flash

At **100 m drone altitude**, the approximate ground distances are:

- **40°**: 84 m
- **60°**: 173 m
- **75°**: 373 m

Mirror configuration:

- Use three **8 × 8 cm** lightweight acrylic or polycarbonate mirrors.
- Approximate total mass: **25–50 g**.
- Keep the rotation axis vertical.
- Mount each mirror at a fixed, adjustable tilt.

- Rotate at approximately **1–3 revolutions per second**.

14.4 Mirror Performance in Bad Weather

Condition	Mirror usefulness
Clear daylight	Excellent
Thin haze	Good
Light rain or snow	Moderate
Dense fog	Poor
Heavy rain or snowstorm	Poor
Sandstorm	Very poor
Thick cloud cover	Unusable

14.5 Smoke Marker

Most emergency flare guns use **red flares**.

Once the drone believes it is near the rescuee, it deploys a fluorescent orange smoke trail to support close-range detection.

Condition	Orange-smoke effectiveness
Clear Dubai desert	Good , although orange can partially blend with sand when viewed against the ground
Snow	Excellent contrast
Light rain	Moderate ; the smoke remains visible but disperses faster
Heavy rain or hail	Poor ; precipitation and wind break up and obscure the plume
Light fog	Limited
Dense fog	Very poor , regardless of smoke color
Sandstorm	Poor ; dust obscures it and strong wind carries it away rapidly
Night	Poor unless illuminated

Rotor wash may destroy the smoke trail unless the outlet is positioned behind and outside the propeller wake.

15. Open Design Decisions Retained from the Source

- Rotating or fixed 360° amber beacon
- Calculated heliograph tilt or a rotating 45–90° tilt range
- LiDAR retained only as an optional sensor